High-Speed In-Person Interior Design Sales Accelerator

Strategic Analysis of Coohom, Sales Closing Psychology, Multi-Modal Room Ingestion & Dual Mobile/Web Architecture

Target Domain: Interior Design & Cabinetry Sales	**Benchmark:** Coohom.com Cloud Suite	**Core KPI:** 15-Minute Consultation-to-Close

Executive Summary: The Consultation-to-Quotation Bottleneck

In the interior design and custom cabinetry industry, traditional sales funnels experience a devastating **40% to 65% drop-off rate** between the initial client consultation and final contract signing. This friction is caused by the multi-day *'Design & Estimation Void'*: a sales rep visits a client's site, takes manual measurements, and retreats to the office for 3 to 7 days to manually build 3D models and calculate the Bill of Quantities (BOQ). By the time the follow-up meeting occurs, client urgency has dropped, competing firms have submitted quotes, and budget anxieties mount.

The Solution: A *Single-Session In-Person Closing Machine*. Equipping sales reps with an interactive tablet and web-synced tool that captures the room, customizes parametric layouts magnetically, calculates a live dynamic budget, and collects a digital signature and deposit in under 15 minutes.

1. Coohom Benchmark: Why Desktop Tools Fail in Live Sales

Coohom is widely celebrated for its robust desktop capabilities (1M+ assets, ExaCloud ray-tracing, parametric cabinetry, and CAD/BIM drawing outputs). However, field testing and practitioner feedback reveal that attempting to use Coohom in front of a live client introduces severe operational failure points:

Dimension	Coohom Desktop Workflow	Sales Accelerator Required UX
Primary Hardware	Desktop workstation, dual monitors, mouse & keyboard.	Touchscreen tablet (Android/iPad) or ultralight laptop on the client's couch.
Catalog Scope	1,000,000+ items. Overwhelming search menus cause choice paralysis.	3 to 5 Curated Style Archetypes (Japandi, Minimalist, Nordic, Industrial).
Object Manipulation	Precision 3D X/Y/Z transformation gizmo. Prone to missed clicks on touch.	Magnetic smart-snapping: units snap flush to adjacent cabinets and wall faces.
Pricing Feedback	Disconnected. Requires manual export to separate quotation spreadsheets.	Live Budget Odometer: Instant real-time CPQ calculation as items or finishes change.
Rendering Latency	Cloud ray-tracing (1 to 5 min wait). Dependent on stable high-speed Wi-Fi.	Instant 60 FPS local WebGL / Three.js. 100% offline-safe in residential basements.
Deal Closing Bridge	Ends with 3D image or panorama link. No on-site commitment step.	On-screen finger signature pad, binding proposal PDF, and Stripe/QR deposit link.

2. Interior Design UX Deep Dive: Bridging Engineer Blind Spots

Software engineers intuitively approach interior design as a 3D geometric modeling problem. However, client consultations are high-stakes sales conversations governed by spatial psychology, loss aversion, and ergonomics:

- **Blind Spot 1: The 'Blank Canvas' Trap:** Giving a client an empty 3D room and asking them to choose cabinets induces paralysis. **Solution:** Curated Style Archetypes. Tapping 'Japandi Warmth' instantly skins the room with matched wall tones, carcass wood, countertop quartz, and hardware.
- **Blind Spot 2: Free 3D Gizmos vs. Magnetic Constraint Snapping:** Touch dragging with 3D gizmos causes cabinets to clip through walls, rotate awkwardly, or float. **Solution:** Magnetic rails. Cabinets are locked to the wall plane and automatically snap flush to neighbors when dragged within 120mm.
- **Blind Spot 3: Spatial Clearances & Ergonomic Safety Rules:** A layout that ignores building codes or workflow ergonomics (e.g. blocking a window, or violating the 900mm kitchen walkway clearance) ruins trust. **Solution:** Visual 'soft collision' guides that highlight clearance thresholds around door swings and worktops.
- **Blind Spot 4: Price Transparency & The 'Budget Odometer':** Hiding the price until the end creates anxiety and sticker shock. **Solution:** A persistent bottom ticker showing Total ($12,450), Monthly ($289/mo), and Deposit ($300). When a finish is upgraded, the odometer animates smoothly.
- **Blind Spot 5: The Closing Bridge (Contract & Deposit):** A great 3D demo without a commitment mechanism results in the client saying 'Thanks, email me the file'—and then ghosting. **Solution:** On-screen finger signing, production slot reservation, and automated WhatsApp proposal dispatch.

3. Multi-Modal Room Input Pipeline

To guarantee zero failure on-site, the sales rep must never be blocked if a floor plan is missing or camera scanning is inconvenient. The system incorporates three robust input tiers:

Modality	Technical Implementation	Best Used When	Speed to 3D
Tier 1: Camera Wall Trace	Snap photo of real wall (eliminates AR drift). Drag obstacle boxes (doors/windows) over frozen frame with real-world mm scaling.	On-site residential visits, single-wall kitchen or wardrobe renovations.	**60 - 90 seconds**
Tier 2: 2D Floor Plan Processing	Ingest architect PDF, real estate brochure, or paper blueprint photo. 2-point scale calibration, vector wall extraction & 3D extrusion.	Client already possesses official developer floor plans or architectural drawings.	**30 - 45 seconds**
Tier 3: 10s Template Preset	Select Single Wall, L-Shape, Galley, or U-Shape room preset. Two sliders: Width (mm) and Ceiling Height (mm).	Client meets in a showroom or coffee shop without blueprints or photos.	**10 seconds**

4. Dual-Platform Architecture: Mobile/Tablet vs. Web Studio

Instead of forcing a single interface to serve both contradictory user personas, the architecture cleanly bisects into two specialized frontends backed by a unified cloud core:

Layer	Mobile / Tablet Sales App (The Closer)	Web Workbench (The Studio Hub)
Target User	Field Sales Representative, Design Consultant.	Senior Interior Designer, Estimator, CNC Factory Admin.
Primary Goal	Fast in-person co-design, live visual quoting, and digital deposit signing.	Master catalog configuration, complex joinery detailing, and CNC DXF export.
Tech Stack	**Flutter + Three.js WebView** (High performance, camera access, offline support).	**React / Next.js + Three.js** (Desktop mouse precision, multi-window layout).

Key Features	• Camera wall tracing & 2D plan import • 1-tap style archetype switching • Magnetic cabinet snapping • Dynamic Budget Odometer • On-screen finger signing & deposit collection	• Supplier catalog & material price matrices • Custom millimeter parametric joinery • Multi-room whole-house project coordination • 4K cloud photorealistic ray-tracing queue • Automated cutting list & factory CNC export
Data Sync	**Lean Scene JSON:** Instant sync via WebSockets / Supabase; full round-trip between tablet and web.	**Unified Backend:** FastAPI pricing engine, PostgreSQL relational DB, and ReportLab PDF generator.

5. End-to-End In-Person Sales Workflow (The 15-Minute Close)

A structured consultative framework designed to guide the customer smoothly from initial measurement to deposit commitment:

Phase	Duration	Sales Action & Customer Experience	System State
1. Space Capture	Min 0 - 3	Snap photo of wall or import client's floorplan PDF. Rep drags door/window boxes. Client sees room digitized in real time.	Parametric 3D wall mesh extruded in local Three.js.
2. Style Co-Creation	Min 3 - 5	Present 4 Curated Archetypes (e.g. Japandi Natural). One tap skins the room. Eliminates catalog overwhelm; establishes aesthetic alignment.	PBR materials applied instantly to scene.
3. Modular Layout	Min 5 - 10	Rep taps 4 to 6 base and wall cabinets. Units snap magnetically along the wall. Client participates by dragging modules.	Collision boundaries enforced; snap guides pulse.
4. Dynamic Quoting	Min 10 - 12	Live Budget Odometer shows itemized total ($12,850) and monthly financing ($310/mo). Rep compares 'Good / Better / Best' material upgrades.	Real-time CPQ engine recalculates line items.
5. The Commitment	Min 12 - 15	Rep taps 'Lock In Design'. 1-page proposal generated. Client signs with finger on tablet to reserve production slot ($250 deposit).	PDF stamped with signature, emailed & texted to client.

Strategic Takeaway: *Traditional platforms like Coohom optimize for the drafting table and rendering farm. Our competitive advantage is optimizing for the consultation couch. By automating snap-alignment, constraining options to curated archetypes, and linking design choices directly to transparent monthly pricing, we eliminate the 7-day quotation void and empower sales representatives to close deals on the spot.*